

HASHING

PROBLEM 5

QUESTION:

DS	Session	Hashing	Question Information	Level 1	Challenge 95
Problem Description You are given a string which comprises of lower case alphabets (a-z), upper case alphabets (A-Z), numbers, (0-9) and special characters like !,-; etc. You are supposed to find out which character occurs the maximum number of times and the number of its occurrence, in the given string. If two characters occur equal number of times, you have to output the character with the lower ASCII value. For example, if your string was: aaaaAAAA, your output would be: A 4, because A has lower ASCII value than a. Input format: The input will contain a string. Output format: You've to output two things which will be separated by a space: i) The character which occurs the maximum number of times. ii) The number of its occurrence. Constraints: The maximum length of the string can be 1000.					
✓ Logical Test Cases Test Case 1 INPUT (STDIN) puppy is a dog!!!!!!!!!!!! EXPECTED OUTPUT ! 12 Test Case 2 INPUT (STDIN) leana is a singer!!!!!!!! EXPECTED OUTPUT ! 7					
✓ Mandatory Test Cases Test Case 1 KEYWORD for(i=0;i<l;i++)					
✓ Complexity Test Cases Test Case 1 CYCLOMATIC COMPLEXITY 6 Test Case 2 TOKEN COUNT 210 Test Case 3 NLOC 35					

SOURCE CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
    char buffer[1005];
    if (!fgets(buffer, sizeof(buffer), stdin)) return 0;

    int str_len = strlen(buffer);
    if (str_len > 0 && buffer[str_len - 1] == '\n') {
        buffer[str_len - 1] = '\0';
        str_len--;
    }

    int occurrences[256] = {0};
    for (int i = 0; i < str_len; i++) {
        occurrences[(unsigned char)buffer[i]]++;
    }

    int max_seen = 0;
    char dominant_char = ' ';
    for (int c = 0; c < 256; c++) {
        if (occurrences[c] > max_seen) {
            max_seen = occurrences[c];
            dominant_char = (char)c;
        }
    }

    printf("%c %d\n", dominant_char, max_seen);
    return 0;
}
```

OUTPUT:

```
leana is a singer!!!!!!  
! 7
```

```
puppy is a dog!!!!!!!!!!  
! 12
```